DESIGN, FABRICATION & COMPARATIVE STUDY OF HYDROPONIC SYSTEM

Project Report submitted to

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

In partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
MECHANICAL ENGINEERING

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CERTIFICATE

This is to certify that the report entitled ***"DESIGN, FABRICATION & COMPARATIVE STUDY OF HYDROPONIC SYSTEM"*** *is a bonafide record of project work done by* ***"Jayakrishnan G. (MUT18ME032), Jen Tom Joseph (MUT18ME033), M Sudarshan (MUT18ME044), Muhammed Inzamam (MUT18ME045)*** *during the year 2021- 2022. This report is submitted to APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Mechanical Engineering.*

Dr. Manoj Kumar K
Guide & Head of the Department
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ABSTRACT

Hydroponics is a type of horticulture and a subset of hydroculture, which is a method of growing plants without soil, by using mineral nutrient solutions in an aqueous solvent. A plant just needs select nutrients, water, and sunlight to grow. Not only do plants grow without soil, they often grow a lot better with the essential nutrients are supplied directly to their roots.

Our main objective is the design, fabrication of a vertical hydroponic farming product and comparative study with A-Frame system, we are recommending two designs, out of which we 3D printed one, we planted spinach as our first crop and compared the crop performance to an A-Frame system.

We analyzed the growth by comparing parameters like crop duration, shoot length, plant spread, root length, root spread nutrient solution consumption, and nutrient content of the crop and we were able to observe that our product performed well in several aspects compared to the A-Frame system.

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LIST OF ABBRIVATIONS

NFT	-	Nutrient Film Technique
LECA	-	Lightweight Expanded Clay Aggregate
pH	-	Potential of Hydrogen
EC	-	Electrical Conductivity
HDPE	-	High Density Polyethylene
LDPE	-	Low Density Polyethylene
PPC	-	Polypropylene Carbonate
PETG	-	Polyethylene Terephthalate Glycol
TDS	-	Total Dissolved Solids
IOT	-	Internet Of Things

CHAPTER 1
INTRODUCTION

Food shortage is one of the modern world's major challenges, and developing ways for sustainable food production has been one of the world's top goals. Uncontrolled deforestation and a growing global population have reduced the amount of arable land available for farming. To deal with this, various approaches were introduced. Hydroponics agriculture is one such approach that was adopted and still stands out as the best suitable method for producing plants without soil. Hydroponics is basically a method of growing plants without soil by using nutrient solution in an aqueous solvent.

Moving on to the history of hydroponics, the man named William Frederick Gericke is said to be the father of Hydroponics. During his period of research in University of California, Berkeley he began to popularize the idea that plants could grow in a solution of nutrients and water instead of soil. He grew 25-foot-high tomato vines just by using nutrients and water. These shocking results made more researchers experiment with the same. It did not take long before the great deal of benefits related to soilless farming was discovered.

1.1 WHY HYDROPONICS?

- Up to 70-90% more efficient use of water.
- Production increases to about 10 times in the same amount of space.
- Many crops can be produced twice as fast in a well-managed hydroponic system.
- Decreasing the time between harvest and consumption increases the nutritional value of the product.
- Indoor farming in a climate-controlled environment means farms can exist in places where weather and soil conditions are not favourable for traditional food production.
- No chemical weed or pest control products are needed when operating a hydroponic system.

1.2 TYPES OF HYDROPONICS SYSTEM

Wick System

It is one of the most basic forms of hydroponics. Here there is a tray consisting of growing media on which the plants are placed. This tray is kept on top of a reservoir containing water and nutrient solution. Wicks are attached to each plant in the tray and the other side is dipped into the reservoir. The nutrient solution travels up the wick to the roots of the plant. The wicks are made of fabric so that the nutrients can easily travel up.

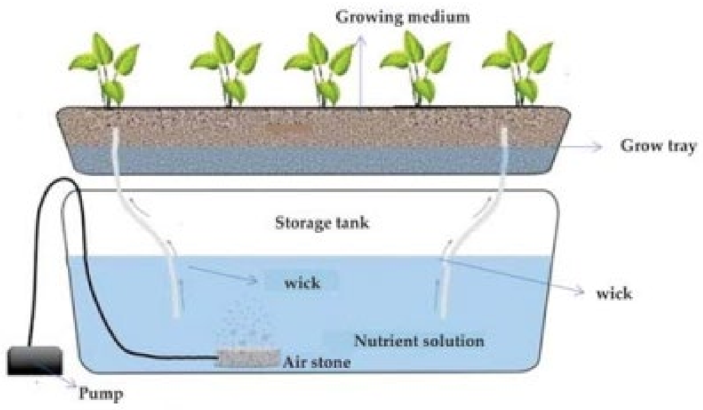


Fig 1.1: Wick system [8]

Deep Water Culture System

It is considered the purest form of hydroponics. Here the plant kept in net pots is suspended over a reservoir containing aerated nutrient rich solution and water such a way that the roots are immersed in the solution.

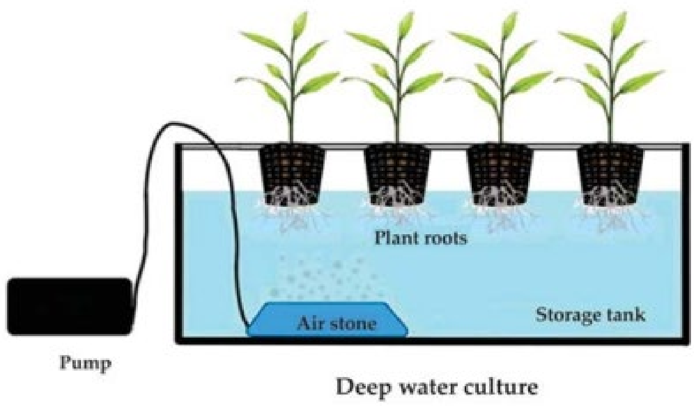


Fig 1.2: Deep water culture [8]

Nutrient Film Technique (NFT)

In this case roots of the plants are suspended in a stream of continuously flowing water containing nutrient rich solution. The channels holding the plants are tilted so that the water containing nutrients can run down and finally fall into the reservoir. Here recirculation of the nutrient rich solution takes place using an air stone.

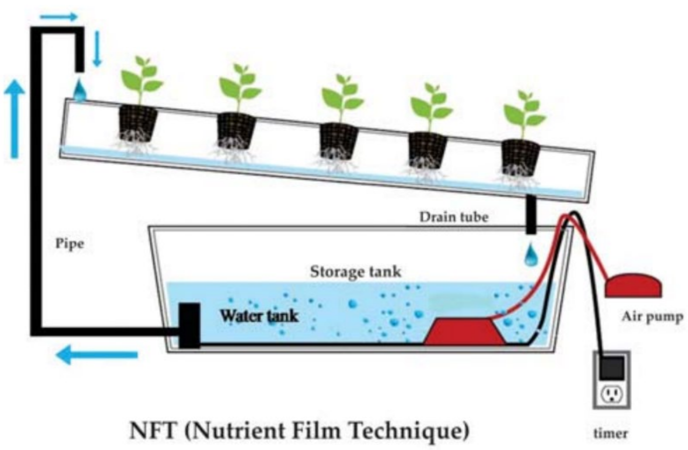


Fig 1.3: NFT system [8]

EBB & Flow System

The plants are placed in large grow beds filled with growing medium and these grow beds are flooded with nutrient solution until it reaches a certain point. A drain makes sure that an overflow does not occur. Power to the pump is controlled by a timer. After a scheduled period of time the pump shuts down allowing the solution to runback through the pump, draining the grow bed completely.

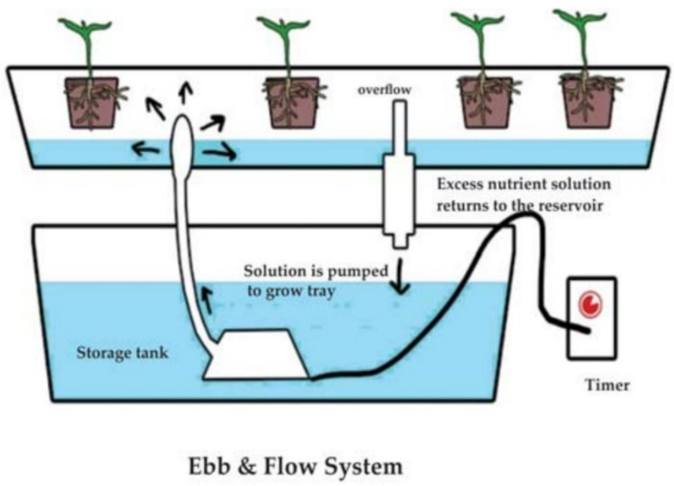


Fig 1.4: Ebb & Flow system [8]

Drip Hydroponics

In the case of these systems nutrient solution is pumped through tubes directly at the roots of the plant. There are adjustable drip emitters at the end of the tubes which allows the nutrient solution to fall drop by drop onto the roots. There are circulatory and non-circulatory drip systems. In the case of non-circulatory drip systems, the nutrient solution falls drop by drop slowly. But in the case of the circulatory system the drops fall faster and the unused solution falls back into the reservoir.

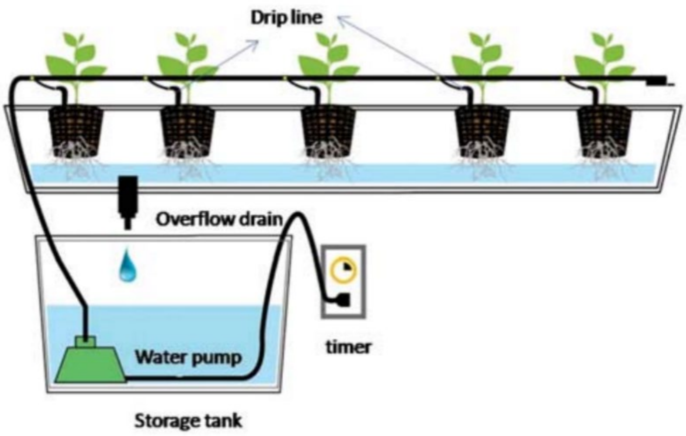


Fig 1.5: Drip hydroponics [8]

Aeroponics

It is a type of soilless farming in which the plants are suspended in air and the nutrient solution is sprayed in the form of mist at the roots of the plants.

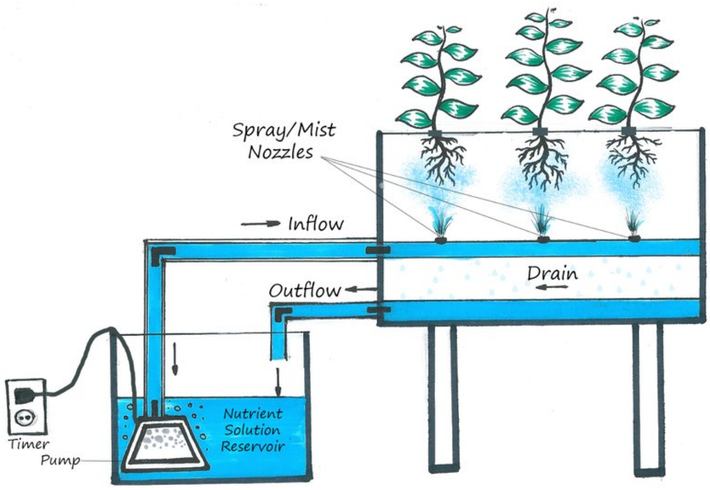


Fig 1.6: Aeroponics
source: https://images.app.goo.gl/FHo1pPfQBrpeNDACA

1.3 WHAT CAN WE GROW?

- Tomato
- Capsicum
- Lettuce
- Brinjal
- Spinach
- Chilly
- Coriander
- Cauliflower
- Beans
- Gourd
- Lady finger
- Sesame seeds
- Mint
- Cucumber
- Radish

Fig 1.7: Spinach source: https://images.app.goo.gl/Mdq6cm

Fig 1.8: Tomatoes Source: https://images.app.goo.gl/iCpPz3HNhsFAf8Sd8

1.4 CROPS OPTIMUM FOR OUR PRODUCT

Tomato

It is an all-season vegetable in India. 3rd on the demand list of vegetables in India with 67%. 2nd on the demand list when the whole world is considered. Hence the best crop to be grown in a hydroponics system.

Lettuce

It is an all-season vegetable in India. 6th on the demand list of vegetables in India with 53%. They have a crop value of 40% in the Indian market.

Spinach

It is an all-season vegetable in India. 8th on the demand list of vegetables in India with almost 50%. They have a crop value of 40% in the Indian market.

Coriander

It is an all-season vegetable in India. It is a very popular medicinal plant that belongs to the Apiaceous family in taxonomic classification, which is widely used as a spice and in pharmacy and in food industries. Due to its multiple uses it is of high demand in India.

1.5 HYDROPONICS MEDIUM

Coconut coir

It is also known as cocopeat, ultrapeat or coco-tek. It is a completely organic medium made from shredded coconut husks having the water retention property of vermiculite and the air retention property of perlite. Finely shredded and steam sterilized coconut husks are the best rooting medium and it protects the roots from diseases and fungus. Also, coir is a completely renewable resource.

Perlite

Perlite is said to be nearly as light as air. Made from air puffed glass pellets, perlite has excellent oxygen retention properties. Its ability to retain oxygen is why it is used as a supplement in soil and soil free mixtures. The main drawback is that due to its light weight the perlite can get easily washed away.

LECA

It stands for Lightweight Expanded Clay Aggregate. It is a coarse growing medium and is made of expanded clay pellets due to its porosity and surface area. These mediums are pH natural and reusable and best suited for hydroponics.

Perfect starts

Recently found, perfect starts are moulded starter sponge made from organic compost and a flexible biodegradable polymer binder. They solve the main problems hydroponic growers face when wanting to use an organic growing medium. They also do not crumble or fall apart during transplanting and help to protect the delicate plant roots. Perfect starts exhibit a perfect air to water holding ratio.

Rockwool

Rockwool is made from molten rock that is spun into long glass like fibres. These fibres are then compressed into bricks and cubes. It has a good water absorbent and drainage property. The main benefit of rockwool is its sterility from pathogens and other things that could contaminate the hydroponics system. Rockwool causes a lot of pollution from its manufacturing; hence its use has come down lately.

1.6 pH IN HYDROPONICS

CROPS	pH
Tomato	6.0-6.5
Spinach	5.5-7.0
Lettuce	6.0-7.0
Cucumber	5.0-5.5
Basil	5.5-6.0
Strawberry	6.0
Celery	6.5
Cabbage	6.5-7.0
Broccoli	6.0-6.8
Parsley	6.0-6.5

Table 1.1: Crops and their recommended pH

The above table shows the optimum value of pH required for different crops. In a nutrient solution pH determines the availability of essential plant elements. The optimum pH range for plants grown in hydroponics is 5.5-6.5, but for some species it can change. Change in pH can cause nutrient imbalance and plants will show some deficiency and toxicity symptoms. Hence it is really important to maintain pH at an optimum level in the hydroponics system.

1.7 HYDROPONICS VS SOIL

Hydroponics	**Soil**
Faster growth	Slow growth
Less susceptible to pests and diseases	More susceptible to pests and diseases
Saves a lot of space (especially while using vertical hydroponics)	More space is required
Less water consumption	More water consumption
Higher yield	Low yield
No soil required	Soil is required
We can use controlled environment and grow plants throughout the year	We cannot grow plants throughout the year
Electricity consumption is high	Less electricity consumption
More maintenance required	Less maintenance
Power outages can kill the crops	No issue with power outages

Table 1.2: Comparison of hydroponics and conventional farming

The above table discusses the comparison between crops grown in soil and crops grown in hydroponics. When plants are cultivated using hydroponics, almost 70-90% less water is required when compared to conventional farming. This is because the nutrient solution can be reused in the case of hydroponics. When crops are grown in soil, the problem of seasonal dependence arises. A particular plant or crop can be grown only during a specific period as it requires a certain environment for perfect growing. But when it comes to indoor hydroponics, we are providing an artificial environment. We can easily control the temperature, humidity and other factors that affect plant growth. Hence crops can be cultivated throughout the year. There is no need for pesticides in hydroponics. The yield obtained by soil cultivation is really low compared to hydroponics. In the case of hydroponics, we are providing the nutrient rich solution directly at the roots, which helps in faster growth and production of higher yield. According to research, crops grown using hydroponics grow about 2 times faster when compared to crops cultivated in soil. Plants grown in soil have a greater risk of attack from pathogens and bacteria. In hydroponics the roots are kept away from soil and there is a lesser risk of pathogen and bacteria attacks. Large amount of space is required for cultivation of crops in soil. Hydroponics can be done in both a vertical and horizontal design. When it comes to using vertical design, the space requirement is reduced significantly. Electricity consumption, maintenance and power outages are the disadvantages of hydroponics. If power supply is not available for too long the plants may die in hydroponics system.

1.8 EASYFARM

Easyfarm is our group name. It will be used before every product designed and manufactured by us.

1.9 LAYOUT OF THE REPORT

The second chapter presents the literature review and the third chapter highlights the objectives of this project. The details of design of all the components used in this study and the elaborate discussion of project methodology is described in chapters four. The results and inferences obtained from the project is presented in chapter five, followed by the conclusion and scope for future work in chapter six. The report ends with the references used for the research and learning about product design and hydroponics.

CHAPTER 2

LITERATURE REVIEW

The research paper "**Evaluation of hydroponic systems for the cultivation of Lettuce (Lactuca sativa L., var. Longifolia) and comparison with protected soil-based cultivation**" [Majid et al 2021] is a comparative study of 2 types of hydroponics systems with that of protected soil-based cultivation. In this study the growth of lettuce in 2 hydroponic techniques, "deep water culture" and "nutrient film technique" was compared to protected soil-based cultivation. This research was conducted in Northern India to identify if hydroponics farming was suitable for growth in India. The results concluded that it is suitable for hydroponic cultivation of lettuce in Northern India. The "deep water culture" and "nutrient film technique" provided superior yield and faster growth compared to soil-based cultivation. According to the paper "deep water culture" is recommended for the growth of lettuce and other leafy vegetables like spinach, mint and basil, due to their simplicity, ease of operation, reduced crop duration, economic factor and their high yield. The "nutrient film technique" on the other hand has a very high water use efficiency and is an environmentally friendly technique. Hydroponic cultivation of leafy vegetables can increase the possibility of reduction of water consumption as well as increase the yield and profitability of production leading to sustainable farming.

The paper "**Vertical farming increases lettuce yield per unit area compared to conventional horizontal hydroponics**" [Touliatos et al 2016] is a comparative study on the growth of lettuce in vertical and horizontal hydroponics systems. It was found out that the vertical hydroponics system can produce about 13.8 times more crop compared to horizontal hydroponics systems. Crop per unit area was more for vertical hydroponics systems. But it was found out that in the case of vertical hydroponics the yield decreased from top to bottom, where as it remained constant in horizontal hydroponics. Also, the shoot fresh weight increased with the increase in light intensity. On the other hand, shoot fresh weight was independent of light intensity in the case of horizontal hydroponics.

According to the journal “**Vertical Farming of Leafy Vegetables under Protected Conditions**” [Jat et al 2019] it was Professor Dickson Despommier in 1999 who came up with a concept of vertical farming. With almost 50% of the Indian population projected to live in cities by 2050, it would not be possible for conventional farming to meet all the needs. This is when vertical farming comes into play. Vertical farming helps to plant more crops per unit area compared to conventional farming and vertical farming using hydroponics is one of the best methods that can be adopted for people living in urban areas who lack land for agriculture. People from urban areas in India are becoming aware of the advantages of vertical farming (especially vertical hydroponics) and have already adopted it for the growth of leafy vegetables like spinach and lettuce.

The journal “**Understanding Hydroponics and Its Scope in India**” [Maiti et al 2020] says that hydroponics farming in India is in the rising stage. The scope of hydroponics in India is really high, with the increase in population the arable land for agriculture is reduced. It is really difficult to meet the needs of the growing population with the help of conventional farming. This is when hydroponics comes into play. Hydroponics does not require soil for growth and also provides higher yield when compared to conventional farming. Hydroponics is also faster and less affected by pesticides, bacteria and weeds. The changing weather conditions in India is not a problem for hydroponics as plants can be grown under controlled environment. Some success stories of hydroponics farming: Aqua Farms, Chennai: Rahul Dhoka, the founder of Green Rusk Organics and Hydroponics Farming Consultancy Aqua Farms grows everything like Italian basil, mint, spinach, lettuce by using planters made of PVC pipes. Letcetra Agritech, Goa: Ajay Naik, software engineer-turned- hydroponics farmer, grows chemical-free vegetables like lettuce, salad greens, cherry tomato, bell peppers etc. Urban Kissan, Hyderabad: Vihari Kanukollu, Dr. Sairam and Srinivas Chaganti established Urban Kissan jointly, where they produce lettuce, herbs, greens, exotic vegetables throughout the year. Future Farms, Chennai: Sriram Gopal, who earlier ran an IT firm, established the Future Farm to develop hydroponics farming, which spread over 10 states growing leafy vegetables.

The paper “**Growth Responses and Root Characteristics of Lettuce Grown in Aeroponics, Hydroponics, and Substrate Culture**” [Li et al 2018] discusses the comparison of growth of lettuce in aeroponics, hydroponics and in conventional farming. According to the results, the lettuce in aeroponics had a greater root/shoot ratio while the lettuce in hydroponics had a higher shoot length than the root. This implies that aeroponics is best suited for plants whose roots are the parts that can be consumed and not the shoot or leaves. But when it comes to hydroponics it was identified that it is best suited for the growth of leafy vegetables like spinach and lettuce.

The research paper “**Modern Practices for the Cultivation of Leaf Vegetables in Hydroponics"** [Kling et al 2020] discusses the results obtained from cultivating different leafy vegetables in vertical and horizontal hydroponics systems. It was found out that lettuce requires about 30-33 days for harvesting in the case of horizontal hydroponics. When it comes to vertical hydroponics lettuce only takes about 27 days for harvesting. When it comes to dill, the harvesting period is about the same in both vertical and horizontal hydroponics. For basil the harvesting period is about 6-10 days earlier in vertical hydroponics. In the case of arugula, parsley and spinach the harvesting occurs about 6-9 days earlier in vertical hydroponics when compared to horizontal ones.

The review paper “**Modern plant cultivation technologies in agriculture under controlled environment: A review on aeroponics**” [Lakhiar et al 2018] outlines a revolutionary technique to plant cultivation in a soil-free environment. Plant roots are suspended in an artificially provided plastic holder, with foam material replacing soil under regulated circumstances in an aeroponics system. The roots are allowed to dangle in the air freely and openly. The nutrient-rich water, on the other hand, is delivered by atomization nozzles. The nozzles produce a fine spray mist with various droplet sizes that can be applied intermittently or continuously. EC & pH values, light & temperature, droplet size affect the growth of plants, proper maintenance of these parameters is essential. According to this study, the aeroponics technique is the best plant-growing technology for food security and long-term development.

The research paper "**Biological functions, uptake and transport of essential nutrients in relation to plant growth**" [Karthika et al 2018] discusses that plant nutrition is concerned with the interplay of soil nutrients and plant growth. This chapter covers the role of nutrients in plant growth and physiology to the greatest extent possible, including information on essential nutrients, their physiological roles, uptake and assimilation, nutritional disorders, the availability of nutrients in soil and their movement to plant roots, and the availability of nutrients to plants via various modes of absorption. Every nutrient is necessary for plants to carry out their physiological tasks, which allows them to grow properly, and its shortage causes specific illnesses.

The paper "**Production of basil (Ocimum basilicum L.) under different soilless cultures**" [Christie et al 2018] is a comparative study of basil grown in hydroponics, aeroponics and peat moss slab. The vegetative parameters, nutrient uptake and oil content were studied. The hydroponics system used was a deep-water culture system. All the results obtained favored the aeroponics system. In the case of hydroponics, the roots were fully immersed in the nutrient solution. But in the case of aeroponics the nutrient solution was directly provided to the roots of the plants in the form of droplets or mist.

The article "**Comparison of Nutritional Content of Spinach (Amaranthus gangeticus L.) Cultivated Hydroponically and Non-Hydroponically**" [Fevria et al 2021] is a comparative study of nutrition content in spinach when cultivated hydroponically and in soil. It was found that the vitamin c content of spinach grown hydroponically (145mg/100g) was less compared to spinach grown in soil (160mg/100g). When the iron content was tested it was found out that both systems had almost same iron content.

2.1 GAPS IDENTIFIED

- Growing food demand, arable land loss and water scarcity due to growing population has increased the need for an efficient farming method.
- The yield in horizontal farming systems is less compared to vertical farming systems.
- Compared to other countries, hydroponic vertical farming with barrels is not widely practiced in India.
- Most of the existing hydroponics systems are bulky and space consuming.

2.2 MOTIVATION

The increasing population has put an immense pressure on conventional farming methods. Due to the increasing population, there is also an increase in demand for food. The loss of arable land and water scarcity also stands in the way of present-day conventional farming. Hence sustainable farming methods like hydroponics and aeroponics have to be adopted to adapt to these challenging circumstances. Also, when it comes to horizontal and vertical hydroponics, the former is much bulkier than the latter.

CHAPTER 3
OBJECTIVES OF THE PROJECT

The objective of the project is to fabricate a modular hydroponic setup for households and compare the crop performance in our vertical barrel system to the A-Frame system, in our project we are proposing two products out of which we fabricated one, we are comparing the crop performance by comparing the vitamin C content, iron content, shoot length, plant spread, root length, root spread, nutrient solution requirements, crop duration, days to maturity and days to harvest.

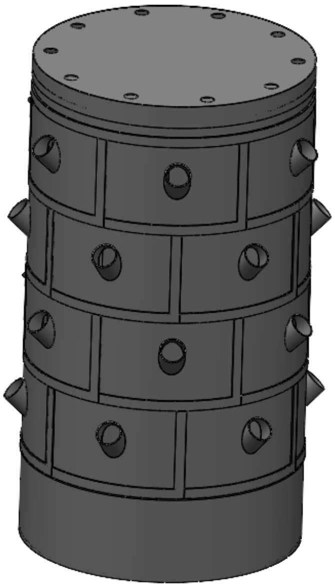


Fig 3.1: Easyfarm vertical barrel 2.0

Fig 3.2: Easyfarm vertical barrel

CHAPTER 4
MATERIALS AND METHODS

4.1 EASYFRAM VERTICAL BARREL 2.0

4.1.1 Material

Material Used: - Polypropylene copolymer

Polypropylene is a thermoplastic that belongs to the polyolefin group of materials. Like most polyolefins, polypropylene is approved for direct food contact. Polypropylene offers a great combination of physical, chemical, mechanical, thermal, and electrical properties, with a good strength to weight ratio. Although polypropylene has lower impact strength than HDPE and LDPE, it is superior when it comes to working temperature and tensile strength. It is lightweight, resistant to staining, and does not absorb moisture. It has excellent resistance to acids, alkalis, organic solvents, and degreasing agents, but has poor resistance to aromatic, aliphatic, and chlorinated solvents. It is also tough, heat-resistant, and semi-rigid, making it ideal for use with hot liquids or gases. It's hard, smooth, surface also makes it ideal for use in areas where bacteria build up is a concern. Polypropylene is available in two basic types either as a homopolymer or a copolymer.

Polypropylene copolymer is a little bit softer and more pliable than the homopolymer but has better impact strength and is more durable. It is also more stress crack resistant and is tougher at lower temperatures. It can run in the temperature range -20º to 180º F. PPC is very versatile and inexpensive. Its lower rigidity is ideal for use in automotive tanks to prevent cracking from road vibration, and orthotic devices. It can also be used for applications that require good chemical resistance and 3-A Dairy compliance. PPC is available in a natural colour; and in the form of rod, sheet, strip, and film. It can be machined with ordinary wood or metalworking equipment.

4.1.2 Parts

The Product consists of the following 4 parts:

1. Segment
2. Bucket
3. Buffer
4. Cap

The Segment

The segment (Fig 4.2) is designed to be manufactured by injection moulding, we have tried to keep the thickness of 2.5mm throughout the design, it will be manufactured with Polypropylene Co-Polymer, the total mass of the part is 190.87 grams.

This cylindrical part projected at 45° out of the calendrical surface will act as a support for the plants which are placed in the segment, the plans will be kept in a pod (a small cup to keep the plant) and the pod will be inserted into the cylindrical opening of the segment, six segments are assembled into a ring, so in one ring we can keep six plants and the assembly of the segments into the ring do not require any fasteners.

The Bucket

The bucket (Fig 4.3) is designed to be manufactured by injection moulding, we have tried to keep the thickness of 3 mm throughout the design, it will be manufactured with Polypropylene Co-Polymer, the total mass of the part is 1872.82 grams.

The bucket can hold up to 50 litre of nutrient solution the ring formed by the assembly of six segments fit on top of this bucket, the submersible pump will be kept in this bucket.

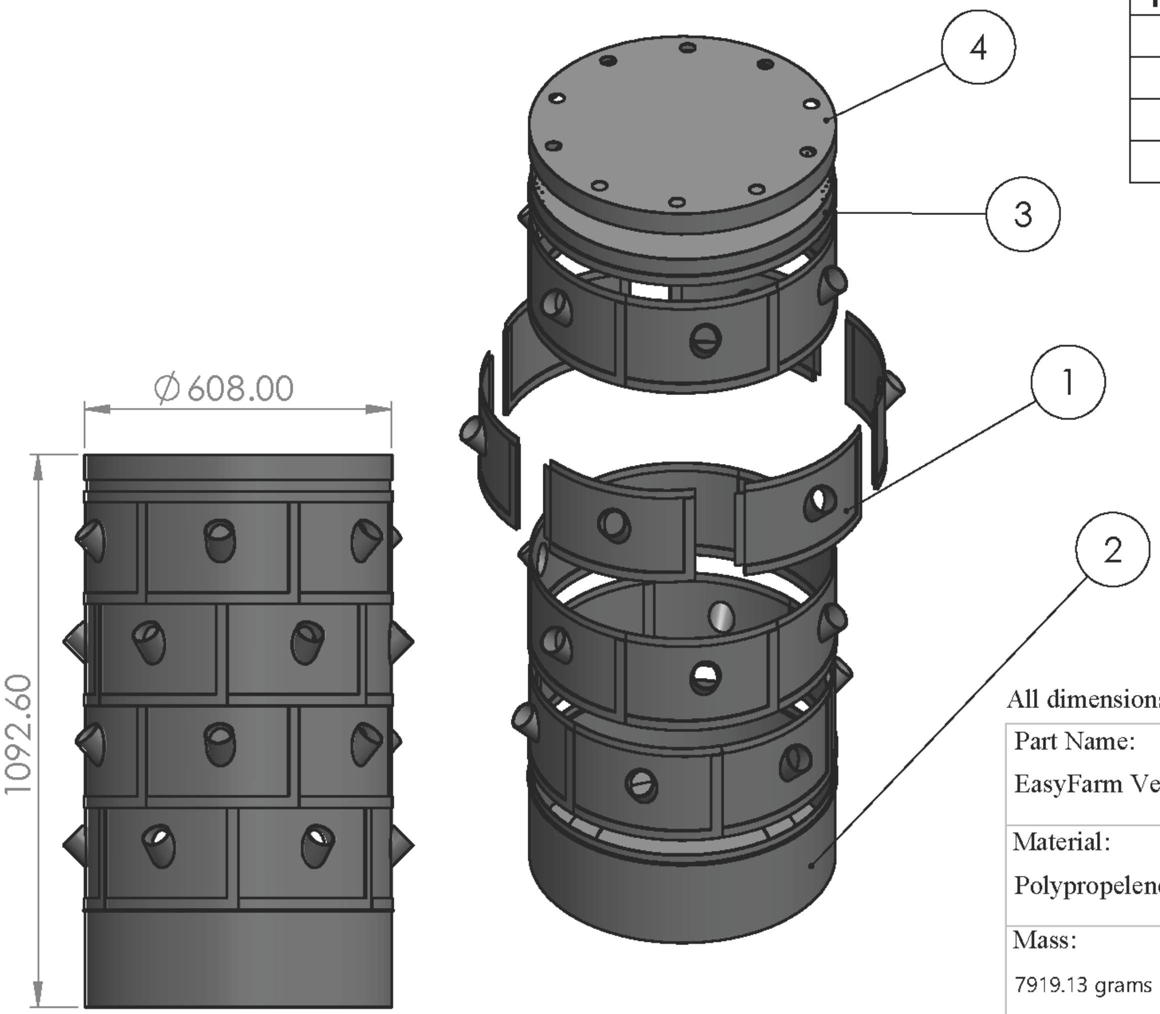
The Buffer

The buffer (Fig 4.4) is designed to be manufactured by injection moulding, we have tried to keep the thickness of 2mm throughout the design, it will be manufactured with Polypropylene Co-Polymer, the total mass of the part is 778.66 grams.The buffer can contain 16 litres before it overflows, the buffer has a hole in the middle for the tube to be inserted and three levels of holes have been given along the circumference of the circular part for letting the solution drip down into the roots of the plants.

The Lid

The cap (Fig 4.5) is designed to be manufactured by injection moulding, we have tried to keep the thickness of 2mm throughout the design, it will be manufactured with Polypropylene Co-Polymer, the total mass of the part is 686.75 grams.

The cap ensures that the water pumped to the buffer doesn’t splatter into the floor.

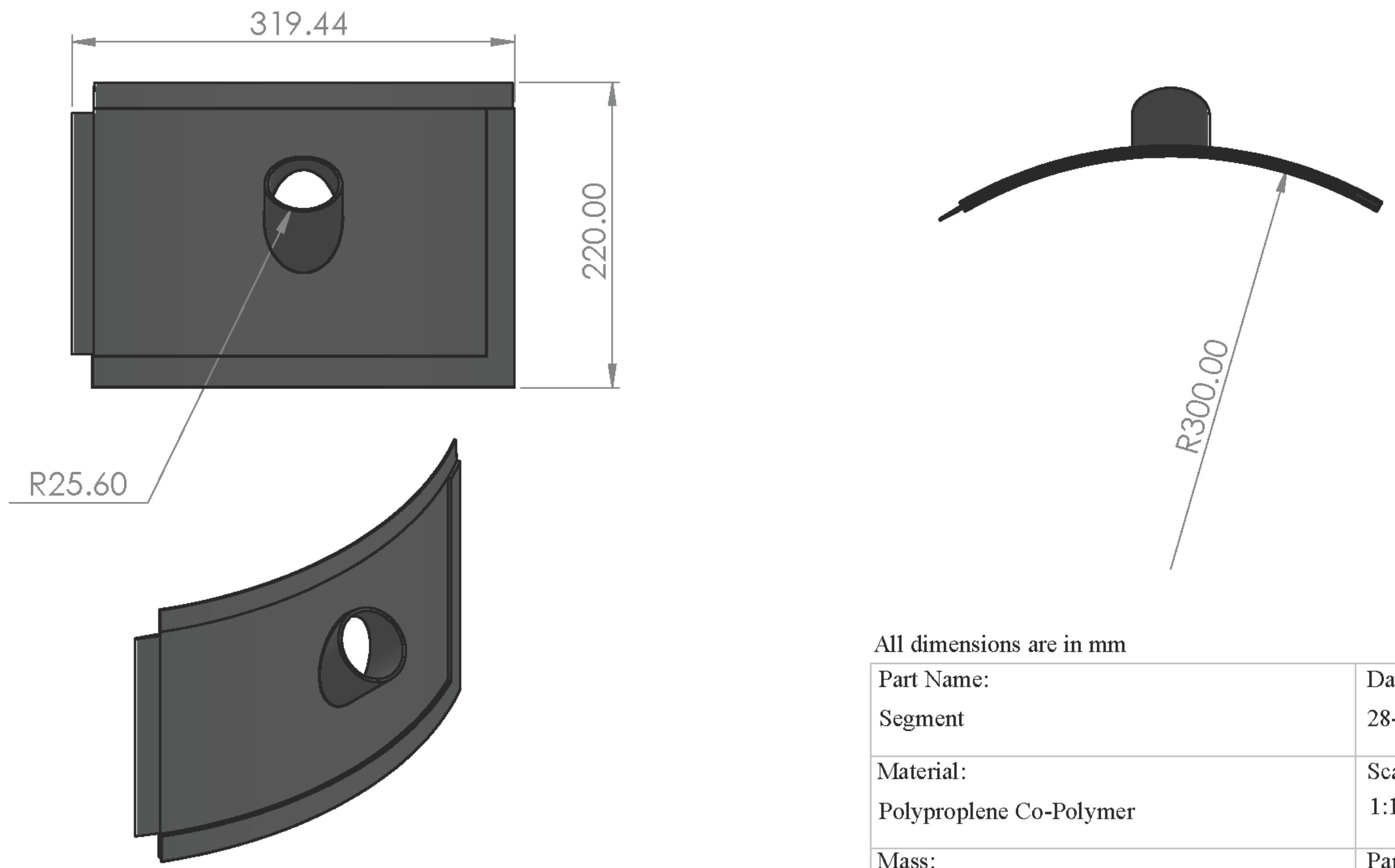


PART NO.	PART NAME	QTY.
1	Segment	24
2	Bucket	1
3	Buffer	1
4	Lid	1

All dimensions are in mm

Part Name: EasyFarm Vertical Barrel Assembly	Date: 01-06-2022
Material: Polypropelene Co-Polymer	Scale: 1:15
Mass: 7919.13 grams	Part no: -

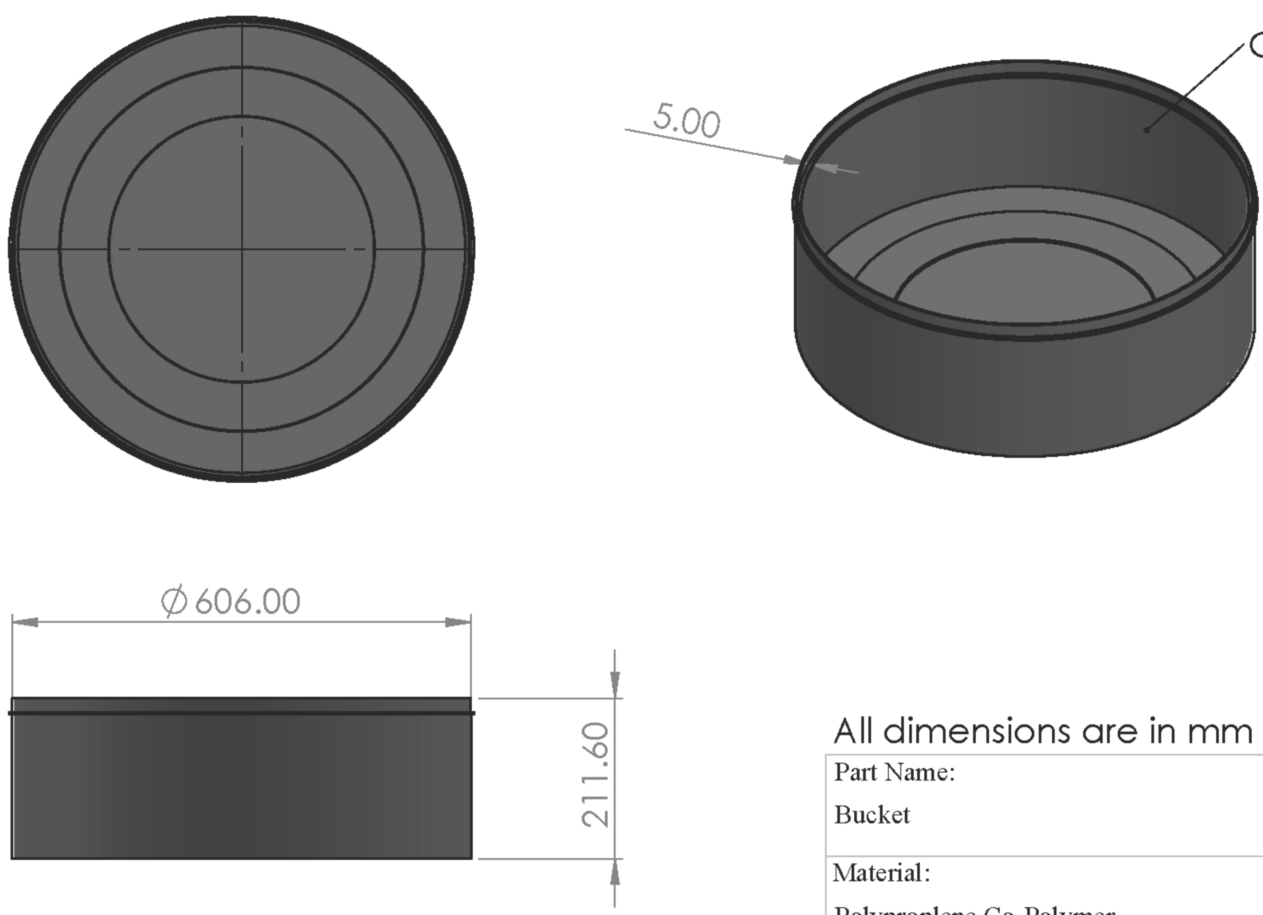
Fig 4.1: Easyfarm Vertical Barrel



All dimensions are in mm

Part Name: Segment	Date: 28-05-2022
Material: Polyproplene Co-Polymer	Scale: 1:10
Mass: 190.87 grams	Part no: 1

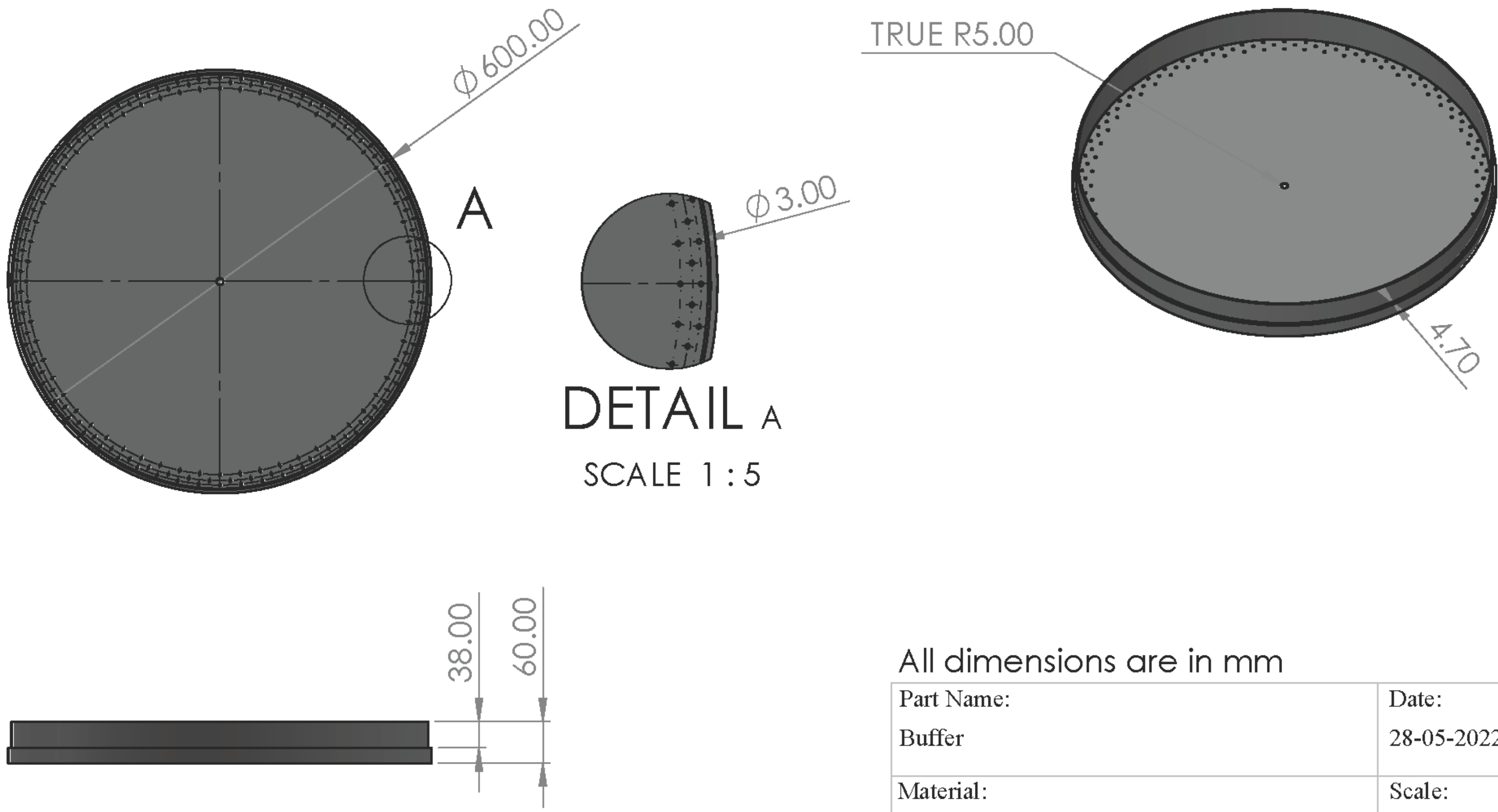
Fig 4.2: Segment



All dimensions are in mm

Part Name: Bucket	Date: 28-05-2022
Material: Polyproplene Co-Polymer	Scale: 1:10
Mass: 1872.84 grams	Part no: 2

Fig 4.3: Bucket



All dimensions are in mm

Part Name: Buffer	Date: 28-05-2022
Material: Polypropelene Co-Polymer	Scale: 1:5
Mass: 778.66 grams	Part no: 3

Fig 4.4: Buffer

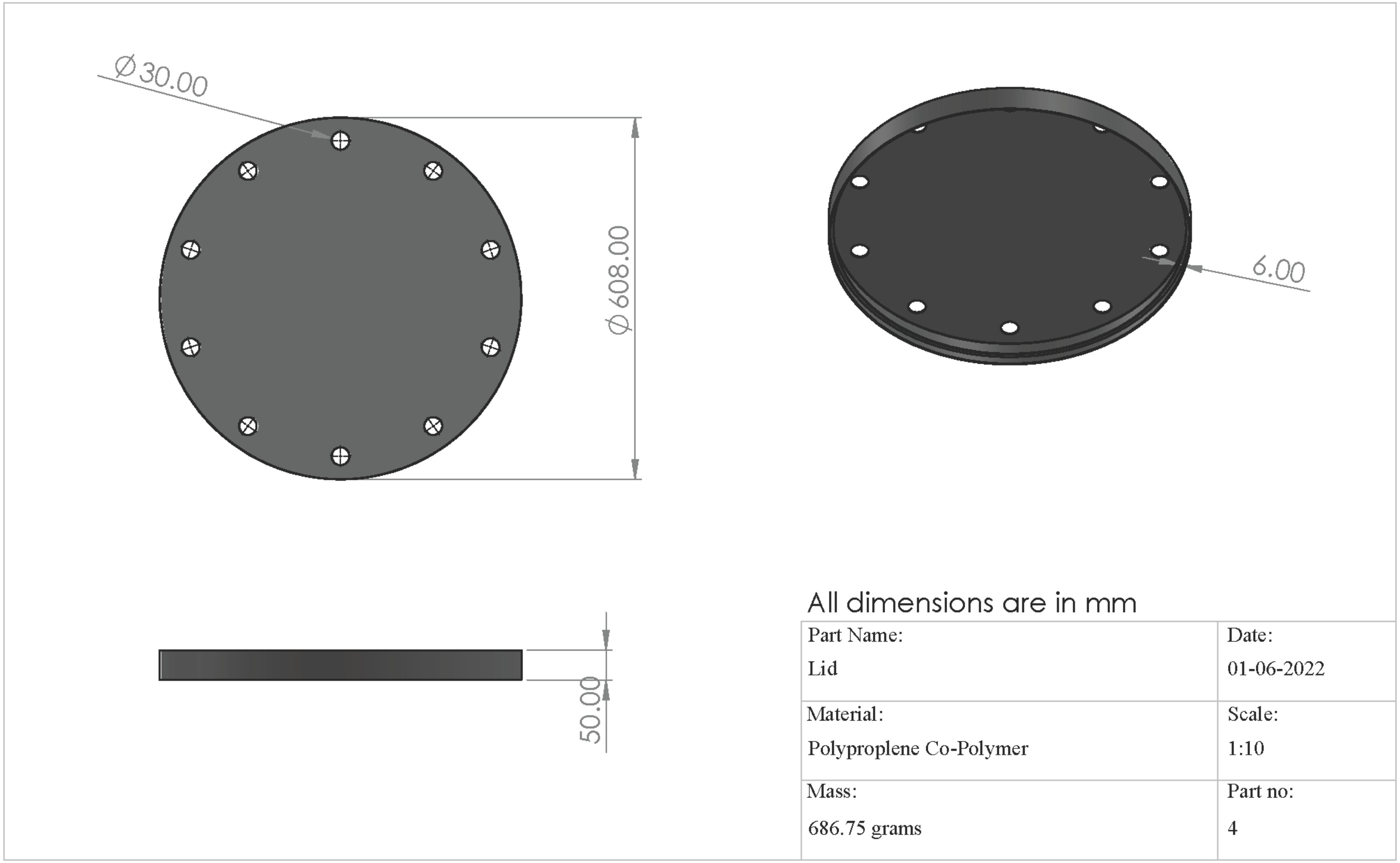

∅30.00
∅608.00
6.00
50.00
All dimensions are in mm
Part Name:
Lid
Date:
01-06-2022
Material:
Polyproplene Co-Polymer
Scale:
1:10
Mass:
686.75 grams
Part no:
4

Fig 4.5: Lid

4.1.3 Working

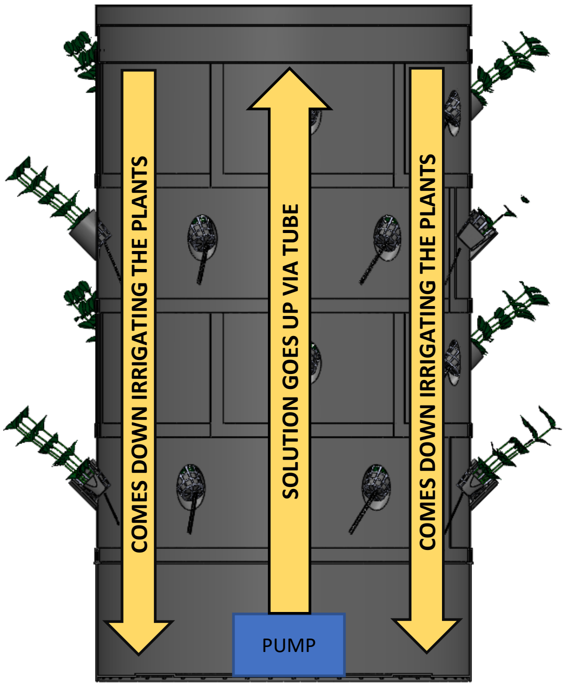


Fig 4.6: Woking diagram

The Fig 4.6 shows a cross sectional view of the complete hydroponic vertical barrel, the submersible pump present in the base bucket pumps the nutrient solution via a tube into the buffer which directs the water towards the roots of the plant.

The flow rate of the pump can be adjusted by using the regulator valve at the end of the tube.

4.2 EASYFARM VERTICAL BARREL

The Easyfarm vertical barrel 2.0 was highly costly to be manufactured by injection molding. Hence, we had to move onto a much smaller product.

4.2.1 Material

Material Used: - PETG

PETG or Polyethylene terephthalate glycol is a thermoplastic polyester commonly used in manufacturing. The PET component is what is commonly found in plastic beverage bottles and food products. The G stands for glycol, which adds durability and strength, and contributes to the compound's impact resistance and ability to withstand high temperatures. To create PETG, glycol is added to PET (Polyethylene Terephthalate) at molecular level. PETG is 100% recyclable and is a food grade material. PETG is also water, chemical and fatigue resistant. The Melting point of PETG is 80°C while its Density is 1.23g/cm^3.

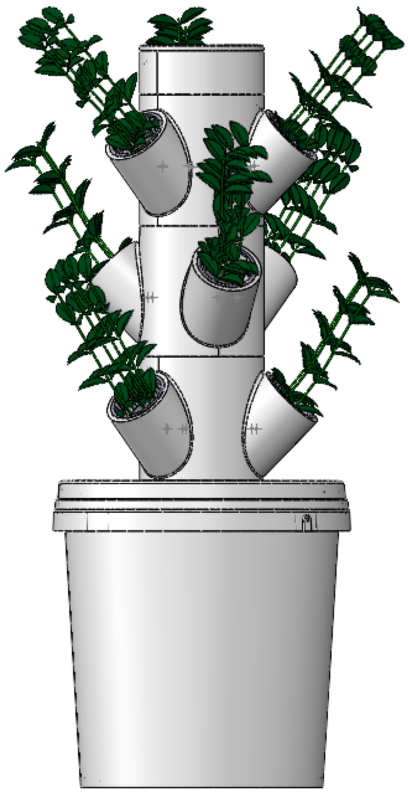


Fig 4.7: Easyfarm vertical barrel

4.2.2 Parts

The 3D Printable design consists of 3 parts which needs to be 3D printed:

- Lid
- Buffer
- Ring

Two parts which is readily available in the market:

- Bucket
- Bucket lid

Lid

The Lid (Fig 4.9) is designed to be manufactured by 3D Printing, we have tried to keep the thickness of 4 mm throughout the design, it will be manufactured with PETG, the total mass of the part is 80 grams.

The lid is designed to fit into the buffer using insert and twist locking mechanism.

Buffer

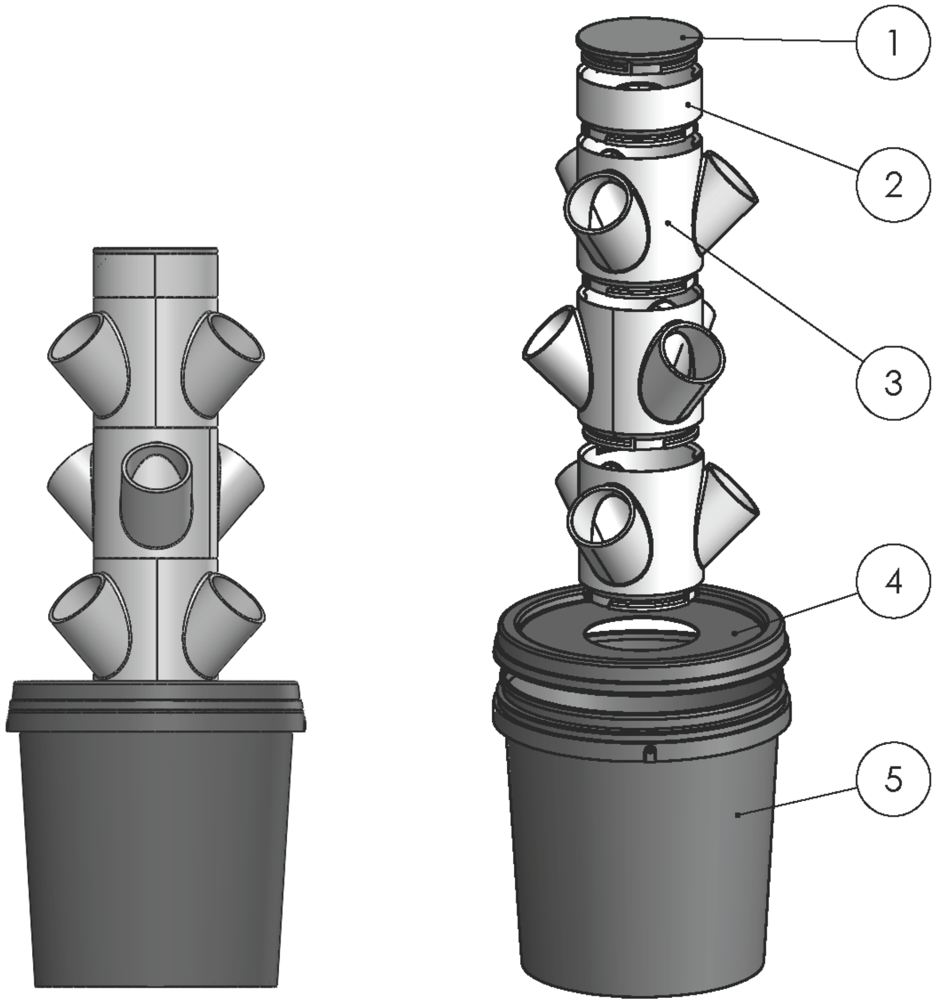
The Lid (Fig 4.10) is designed to be manufactured by 3D Printing, we have tried to keep the thickness of 4 mm throughout the design, it will be manufactured with PETG, the total mass of the part is 115.93 grams.

The buffer can contain 500ml before it overflows, the buffer has a hole in the middle for the tube to be inserted and three levels of holes have been given along the circumference of the circular part for letting the solution drip down into the roots of the plants.

The Segment

The segment (Fig 4.11) is designed to be manufactured by 3D printing, we have tried to keep the thickness of 4 mm throughout the design, it will be manufactured with PETG, the total mass of the part is 260 grams.

These cylindrical parts projected at 45° out of the calendrical surface will act as a support for the plants which are placed in the segment, the plans will be kept in a pod (a small cup to keep the plant) and the pod will be inserted into the cylindrical opening of the segment, there are 3 pods in a ring, so in one ring we can keep three plants and the assembly of the ring one over the other do not require any fasteners.

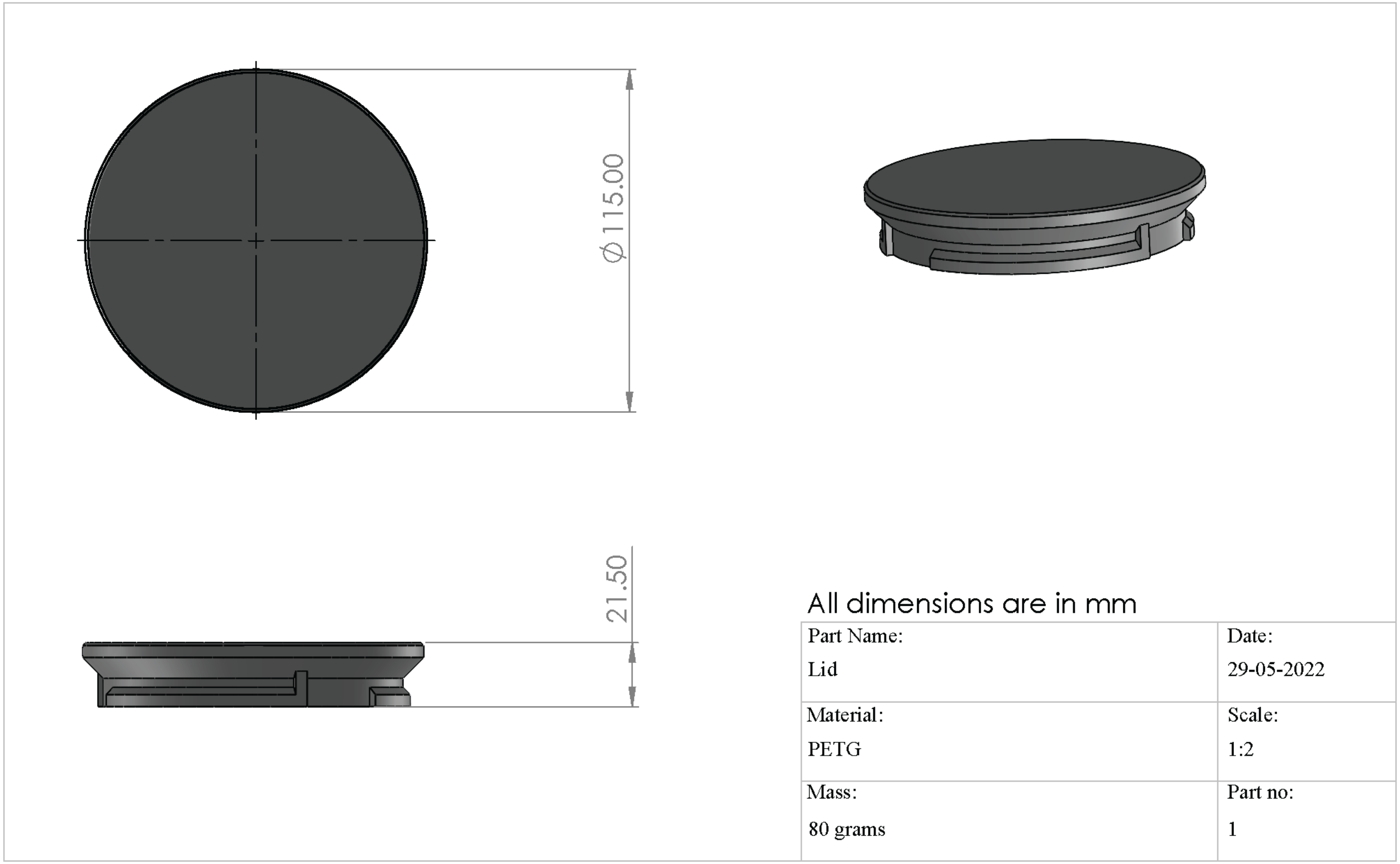


PART NO.	PART NAME	QTY.
1	LID	1
2	BUFFER	1
3	RING	3
4	BUCKET LID	1
5	BUCKET	1

All dimensions are in mm

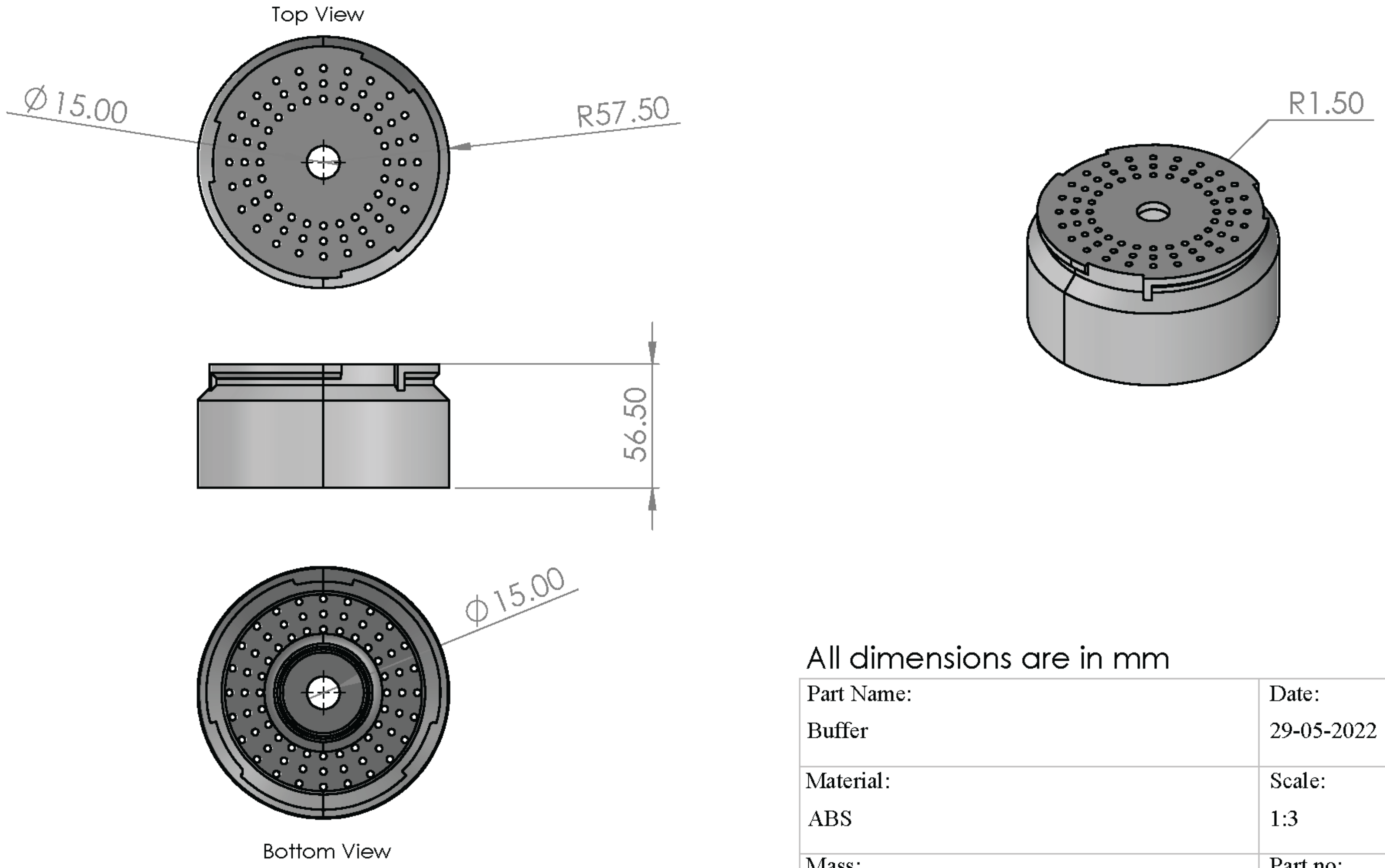
Part Name: 3D Printed vertical barrel	Date: 30-05-2022
Material: PETG	Scale: 1:7
Mass: 1451.65 grams	Part no: -

Fig 4.8: 3D Printable Design Assembly



Ø115.00
21.50
All dimensions are in mm
Part Name:
Lid
Date:
29-05-2022
Material:
PETG
Scale:
1:2
Mass:
80 grams
Part no:
1

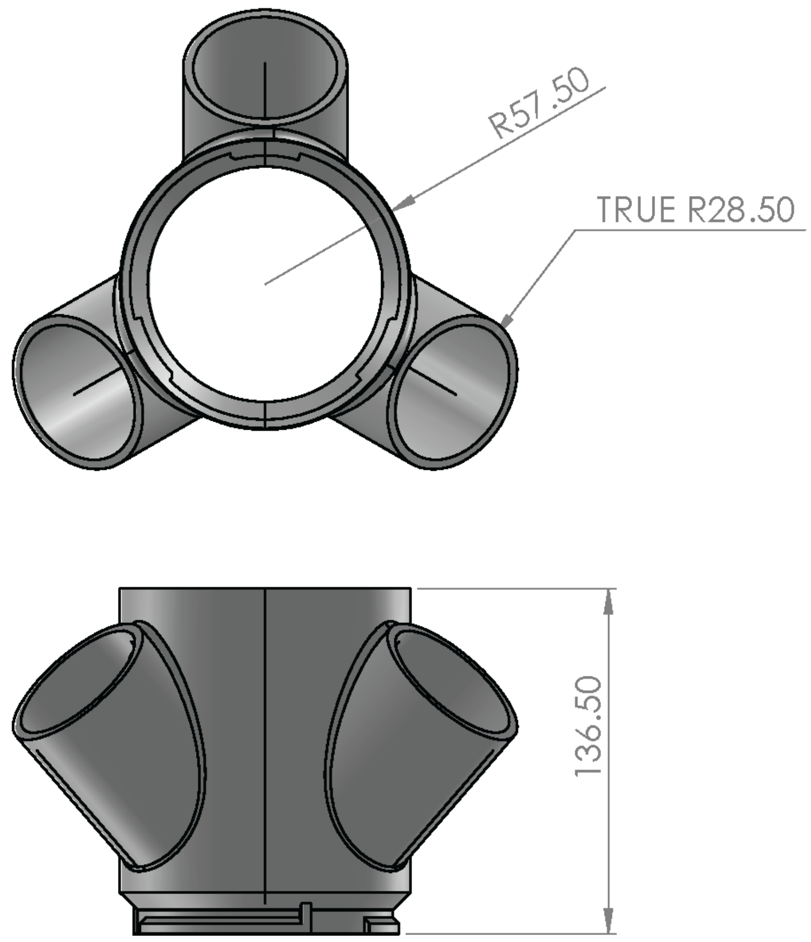
Fig 4.9: Lid

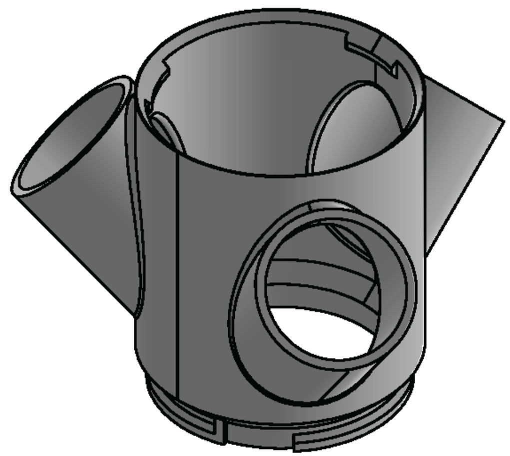


All dimensions are in mm

Part Name: Buffer	Date: 29-05-2022
Material: ABS	Scale: 1:3
Mass: 115.93	Part no: 2

Fig 4.10: Buffer

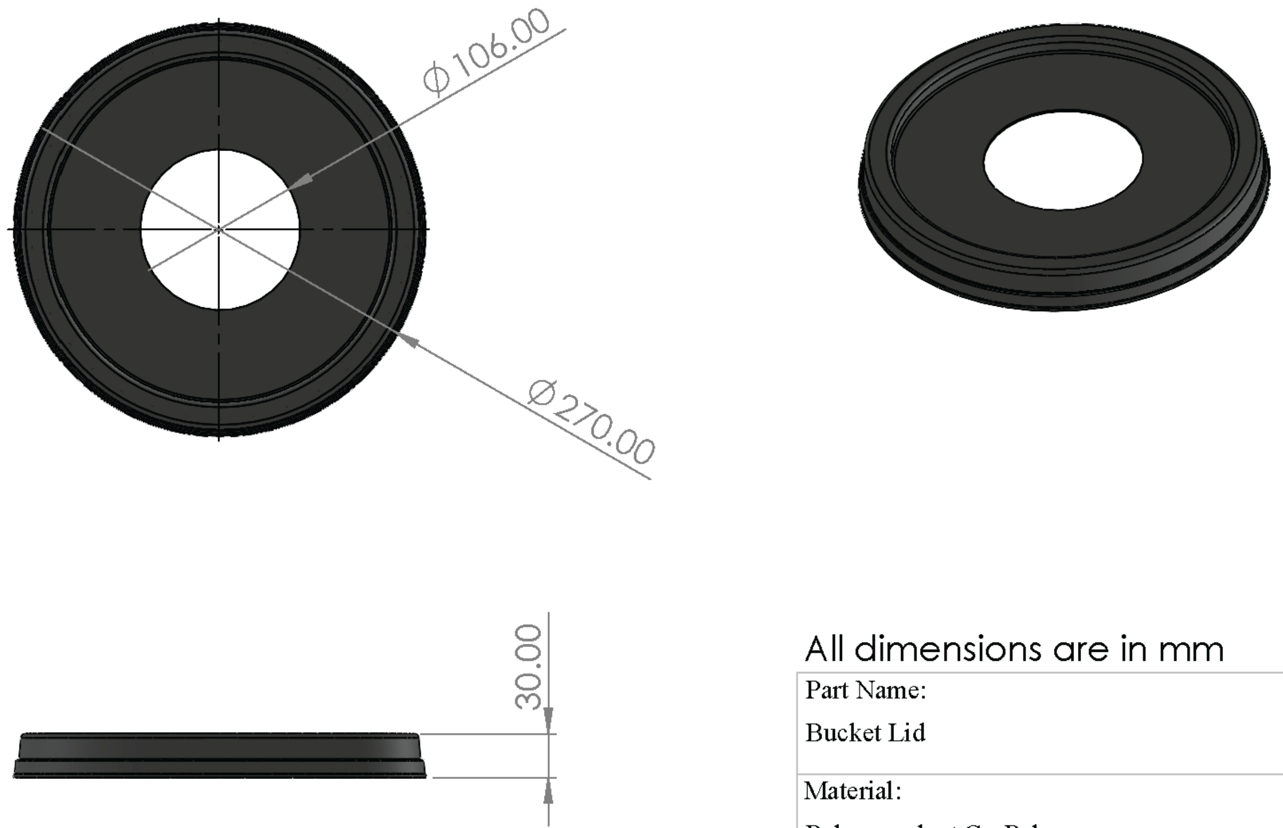




All dimensions are in mm

Part Name: Ring	Date: 29-05-2022
Material: PETG	Scale: 1:3
Mass: 259.43	Part no: 3

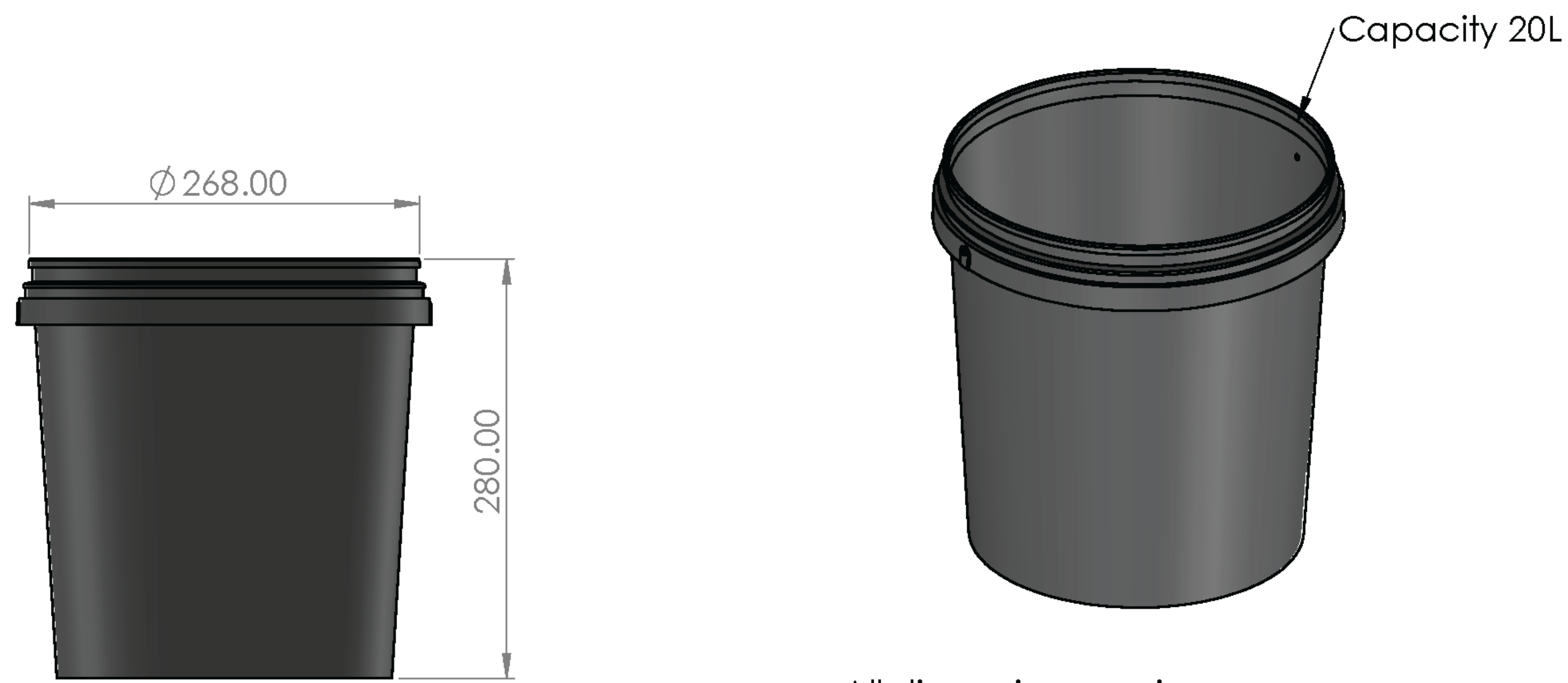
Fig 4.11: Ring



All dimensions are in mm

Part Name: Bucket Lid	Date: 29-05-2022
Material: Polypropylent Co-Polymer	Scale: 1:5
Mass: 114.79 grams	Part no: 4

Fig 4.12: Bucket Cap



All dimensions are in mm

Part Name: Bucket	Date: 29-05-2022
Material: Polypropylene Co-Polymer	Scale: 1:5
Mass: 362 grams	Part no: 5

Fig 4.13: Bucket

4.2.3 Working

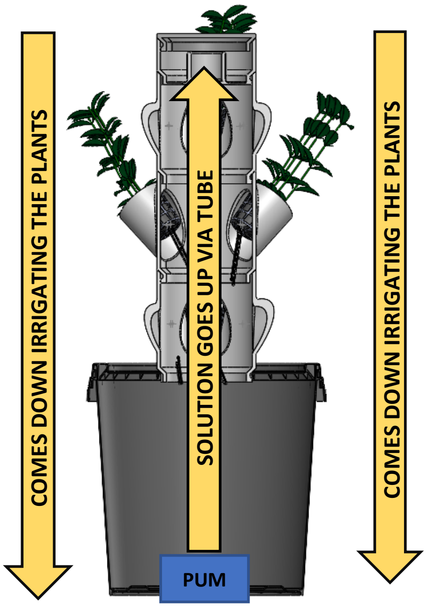


Fig 4.14: Woking diagram

The Fig 4.14 shows a cross sectional view of the complete hydroponic vertical barrel, the submersible pump present in the base bucket pumps the nutrient solution via a tube into the buffer which directs the water towards the roots of the plant.

The flow rate of the pump can be adjusted by using the regulator valve at the end of the tube.

4.3 EXTRA ITEMS

Nutrient Solution

The nutrient solution was brought from PlantMe Agro Solutions Pvt Ltd, there are two types of nutrient solutions named nutrient A and nutrient B, nutrient A is a plant growth booster, it is an organic growth promoter which enhances the natural immunity, flowering, fruiting and foliage of the plant. It is a 100% eco-friendly nutrient and can be used for all agricultural and horticultural products in all the stages of the plant the ingredients of nutrient A are chitosan gel and chelated forms of minerals like Boron, Calcium, Cobalt, Iron, Magnesium, Manganese, Molybdenum, Zinc. Nutrient B is a unique blend of micro and macro elements along with accurate addition of trace elements which are well balanced and act as an aid for growth of the plant

Pump

The pump is a quiet submersible pump of power 40W and a flow capacity of 1500 liter per hour

TDS Meter

Used to find the Total Dissolved Solids, helps in monitoring weather enough nutrients are available in the nutrient solution

pH Meter

Used to find the pH of the nutrient solution, we have to maintain a pH level between 5.5 to 7, low pH can indicate magnesium deficiency or Iron and Aluminum toxicity. High pH levels can prevent uptake of Calcium and Phosphorous.

A- frame system

Fig 4.15: A- frame system

The A- frame system is an already existing hydroponic system. The Easyfarm vertical barrel is compared with this system.

4.4 SITE SELECTION

A perfect place where enough sunlight and also protection from rain is necessary. Hence the Hydroponic system was placed in the roof of M George Block at Muthoot Institute of Technology and Science.

4.5 SEEDLING PRODUCTION

Seeds were purchased from Plantme Agro Solutions Pvt. Ltd and germinated at the college before transferring it into the system for further growth.

4.6 PLANT SELECTED

It was identified that hydroponics is more suitable for leafy plants, therefore spinach was selected as the crop for our first cropping cycle

4.7 COST

Description	**Cost in Rupees**
3D Printed Vertical Barrel	13000
A-Frame system	10730
Nutrient Solution	352*3=650
Pump	644*2=1288
pH Meter	1900
Green Amaranthus seeds	20
Clay Pellets	100
Reservoir	180
Hose	50
Total Cost	**27918**

Table 4.1: Total cost table

4.8 GROWING METHOD

The seeds were kept in the moist dark area to germinate them. Once the seedlings were at least 2cm tall, they were transferred to the hydroponic growing system. To ensure that the reservoir had enough nutrients a TDS meter was used. To ensure that the pH of the nutrient solution was optimum pH meter was used. The pH and TDS were monitored every day to ensure that plants were getting all the necessary nutrients required for their growth. When the TDS was low nutrient solution was added to adjust it. When pH got low sodium bicarbonate was used to increase the pH.

CHAPTER 5
RESULTS AND DISCUSSIONS

5.1 CROP PERFORMANCE

Growth parameters were recorded to analyse the effect of different systems on plant growth. Crop performance was monitored by planting the same crop in two different systems with the same working technique. Growth systems had a considerable influence on the performance of the plants in both of the two growing systems studied. Crop duration, crop growth, crop quality and yield were measured to conclude the results.

5.1.1 Crop duration

Crop duration includes the details of the crop from planting seeds, days of transplant, days to mature and days to harvesting. These details are provided in the table below. The days of transplanting and harvesting were similar for both the growing systems as they were raised together in the same manner and cultivated at definite time.

System	Days to transplant	Days to maturity	Days to harvest
A-frame system	7	14	17
Easyfarm Vertical Barrel system	7	15	17

Table 5.1: Crop duration

Fig 5.1: Plants at mature stage

Fig 5.2: Plants at harvesting stage

The results of crop duration were comparatively similar in both the systems even though the A-frame system reached the mature stage quicker due to the increase in nutrient absorption.

5.1.2 Crop growth

The study showed a noticeable difference in the impact of growth parameters on plants grown on different systems. Despite the fact that the pH and conductivity of the nutrient solution were constantly changing, hydroponic plants demonstrated exceptional growth. Design of the system, flow of water, had a clear influence on the crop growth. The details of root and shoot growth parameters is discussed below in the table.

Parameters	Easyfarm vertical barrel system	A-Frame system
Shoot length	23 cm	26 cm
Plant spread	High with more leaves	High with more shoot
Root length	Moderate	Moderate
Root spread	Long and slender	Short and wide

Table 5.2: Crop growth comparison

The plants grown in the A-Frame system had a comparatively large shoot length, because the plant is perpendicular to the ground plane in the A-Frame system compared to a 45° in the vertical barrel system, this also led to the bending of plants. Growing pods contain the same number of plants in all systems, but when they grew up the externally bought system had a dominating plant along with the others being of similar growth (not matured). In the case of vertical barrel one to two plants were dominating and the others didn't complete the juvenile stage. As a result, the A-frame system had thicker vegetation but the other one had more leaves.

Considering the growth characteristics of roots in different systems, it was totally different. The vertical system has the nutrient solution flowing over the roots and its vertically downwards due to the system design. Consequently, the roots grow downwards along the flow, as the flow being in the form of drip the roots were long and slender. In the case of the A-Frame system the roots were spread entirely through the pods, reason being the roots are immersed in the nutrient solution and the flow is perpendicular to the root growth allowing it to spread throughout the water.

Fig 5.3: Root Comparison

Fig 5.4: Shoot in Easyfarm vertical barrel

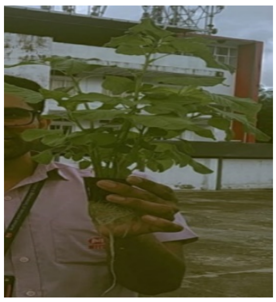


Fig 5.5: Shoot in A-Frame system

5.1.3 Total Dissolved Solids

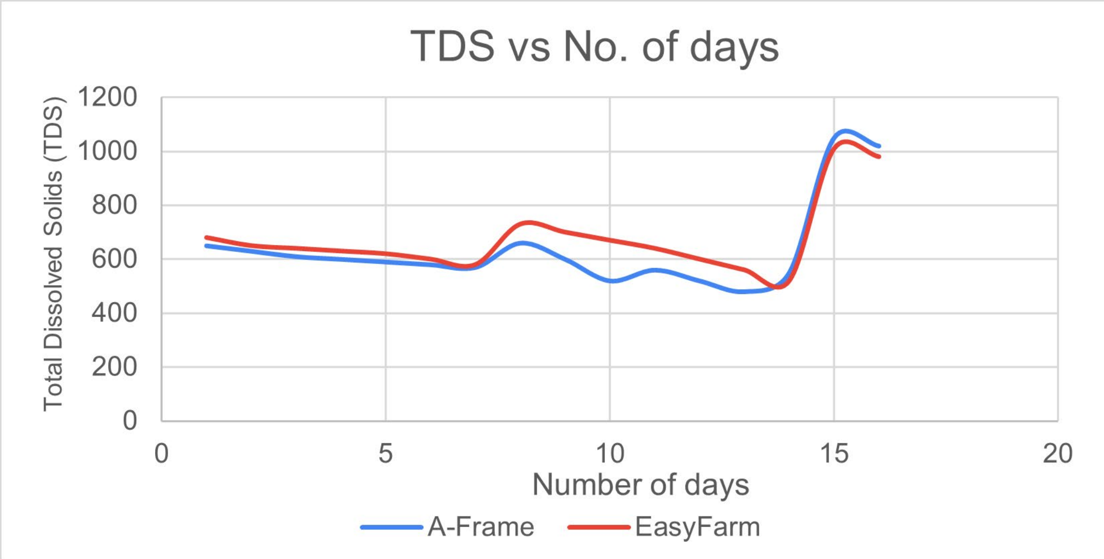


Fig 5.6: TDS vs no of days Plot

The TDS (Total Dissolved Solids) of both the systems was daily monitored and the obtained data was plotted against the number of days. Both the systems were regularly monitored and the required amount of nutrient solution was added to it as needed by the plants. The graph shows the exact variation of TDS level of both the systems.

At first the TDS level was maintained between 500-650 and there was a regular decrease in the TDS level for the first week as seen in the graph. After one-week nutrient solution was added and the TDS level of about 800 was maintained. As seen in the graph there was a larger dip in the TDS level as the plants grew and they were in high demand for nutrients. However, when the Graphs for both the systems are closely analysed there was a larger dip for the system using the A-frame design. This is because in the A-frame system the roots are always being submerged in the nutrient solution and there will be more absorption of nutrients from the solution. But in the case of vertical barrel system the solution is being dripped on the plant roots and the plants get only the needed nutrients. An extra amount of nutrient solution needed to be added to the A-Frame system to cope up with the demands of the plants. But the plants when they absorb more nutrients than they need it's just a waste of the solution as the plants cannot use this for the preparation of food. After 2 weeks nutrient solutions were again added and a TDS level around 1000 was maintained. After that when the plants reached maturity, the plants were harvested.

5.1.4 Crop quality

To check the crop quality, vitamin c content and iron content of spinach grown in both the systems were tested and compared. UV spectrophotometer was used for iron content test. To find the vitamin c content titration method was used. It was found out that vitamin c content in spinach grown in Easyfarm vertical barrel was more compared to the A- frame system. The iron content was slightly more in the case of spinach grown in A- frame system.

SYSTEM	Easyfarm vertical barrel	A- frame system
VITAMIN C (mg/100g)	76.40	57.33
IRON (mg/100g)	0.074	0.078

Table 5.3: Crop quality (Vitamin c and Iron)

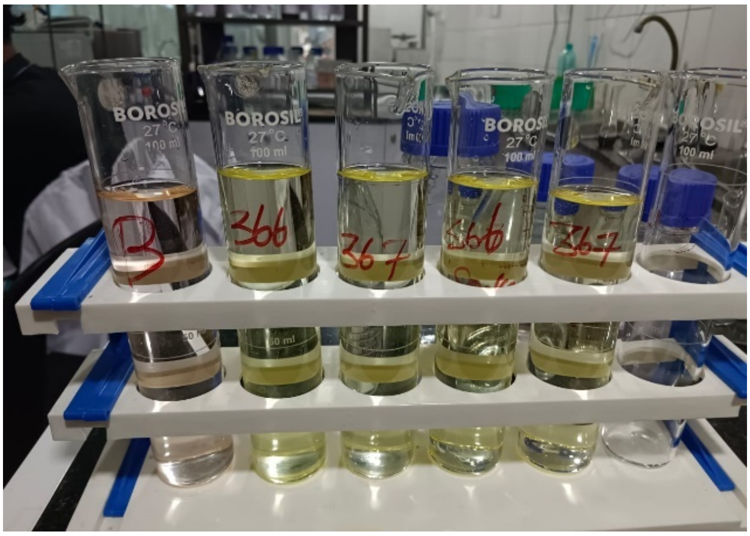


Fig 5.7: Spinach test samples

5.2 FINDINGS

1. Water consumption in the vertical barrel system was comparatively low with the A- frame system.

   Reasons for high water consumption in A- frame system; -

   - Larger surface area for water to evaporate.
   - Roots being immersed in water consume more water.

   Reasons for low water consumption in vertical barrel system: -

   - Nutrients supplied by flowing water continuously drop over the roots of plants.
   - No stagnant water surface present for evaporation.
2. Clogging of nutrient solution found in A-frame system due to wider roots.
3. Smaller diameter of the vertical barrel system caused a lack of space for the plants to grow and flourish.
4. Disease detection was found to be more in the A-frame system due to the stagnant water present.
5. Cleaning of the vertical barrel system after one crop cycle was found to be easier than the A-frame system.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

A vertical barrel hydroponics system was designed and fabricated by 3D printing. It was then compared with an A- frame system. Spinach was cultivated in both systems and parameters like crop growth, crop quality, TDS etc were compared. The plants grown in the A-Frame system had a comparatively large shoot length. the A-frame system had thicker vegetation but the vertical barrel had more leaves. In the case of vertical barrel system the roots were slender and long. But in the case of A- frame system the roots were short and wide. vitamin c content in spinach grown in easyfarm vertical barrel was more compared to the A- frame system. The iron content was slightly more in the case of spinach grown in A- frame system. Water can reach all the plants in vertical barrel systems, whereas in A frame system some plants may not get water due to clogging of pipe by the wide roots. Water consumption in the vertical barrel system was comparatively lower than the A-frame system. Cleaning of vertical barrel system was easier than cleaning of A- frame system due to the easy segmented design of easy farm vertical barrel. The smaller diameter of vertical barrel system caused a lack of space for the plants to grow and flourish.

The Easyfarm vertical barrel has a lot more to offer. The Easyfarm vertical barrel 2.0 will be the main focus for the future. Vertical barrel system can be modified such that it can be easily converted to aeroponic systems. Development of our own nutrient solution is another future plan. A locking mechanism for pods can be implemented to make it stable. Methods have to be found out to manage issues related to power outage. The best plant for the easyfarm vertical barrel can be found out by trial-and-error method. Last but not the least, integration of IOT into the easyfarm vertical barrel system to make it more efficient and productive.

CHAPTER 7

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